AMAN SINGH PATEL

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CAREER OBJECTIVE

AI/ML Engineering student skilled in Machine Learning, Deep Learning, Generative AI, and Agentic AI. Experienced in building LLM-powered applications, multi-agent systems, and RAG pipelines using Python and modern AI frameworks.

EDUCATION

Pranveer Singh Institute of Technology 2023 – 2027
B.Tech in Artificial Intelligence and Machine learning (AIML) -70.35% Kanpur, Uttar Pradesh

St. Xavier's High School 2023
Class 12th -67% Banda, Uttar Pradesh

TECHNICAL SKILLS

Programming: Python, SQL, C/C++
AI/ML: Machine Learning, Deep Learning, Transformer Architecture, PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib
Generative & Agentic AI: LLMs, RAG, AI Agents, Multi-Agent Systems, Agent Orchestration, Prompt Engineering, LangChain, LangGraph, Hugging Face, Ollama, MCP
NLP: Embeddings, Tokenization, Text Classification, Information Retrieval
Backend & APIs: FastAPI, Pydantic, REST APIs, JWT, OAuth2
Databases: PostgreSQL, MySQL, Qdrant, ChromaDB
Core Subjects: OOP, DBMS, Operating Systems, Computer Networks
Tools: Git, GitHub, Jupyter Notebook, Google Colab, VS Code

PROJECTS

AI-Native Adaptive Learning Engine 2026
Python, FastAPI, LLMs, RAG, PostgreSQL, Qdrant, MCP

- Designed an AI-native adaptive learning engine that continuously personalizes exam preparation by modeling exam requirements from syllabus/PYQs and learner state from mock performance, doubts, and interaction history.
- Built a multi-agent architecture with a central orchestrator that delegates tasks to specialized agents for exam analysis, strategy generation and refinement, mock generation, and context-aware QA workflows.
- Implemented targeted RAG pipelines for study content, PYQs, and learning memory, with PostgreSQL for structured learner state and MCP-based controlled access to knowledge and reports.
- Developed an adaptive feedback loop that converts mock results, chat sessions, and recurring misconceptions into learning events and automatically reprioritizes the learner's study strategy.

RAGify India – Legal AI Research System April 2025
Python, LangChain, Mistral-7B, FAISS, HuggingFace

- Developed a Retrieval-Augmented Generation (RAG) system using Mistral-7B, LLaMA, LangChain, FAISS, and semantic search for legal question answering across 50+ Indian legal acts.
- Designed a scalable document ingestion pipeline involving preprocessing, chunking, embedding generation, FAISS indexing, retrieval optimization, and end-to-end AI pipeline development.
- Constructed a benchmark dataset of 500+ legal queries for retrieval benchmarking, model evaluation, experimentation, and generation quality analysis.

AI Trip Planner (Team Project) Nov 2025
Python, LangChain, Multi-Agent LLMs, OOP

- Developed a multi-agent AI trip planning system in Python consisting of Research, Optimization, and Recommendation agents for personalized itinerary generation.
- Designed a weighted optimization algorithm combining cost, travel time, ratings, and user preferences to generate personalized and constraint-aware travel itineraries.
- Built automated context-aware personalization using travel season, regional preferences, and destination metadata.

CERTIFICATIONS

- Supervised Machine Learning: Regression and Classification – DeepLearning.AI
- Python – Infosys Springboard
- Cloud Infrastructure 2025 AI Foundations Associate – Oracle

APPLIED RESEARCH PROJECT

- **Legal AI System Analysis (RAG Chunking Comparison):** Evaluated different text chunking techniques across Mistral-7B, Llama3-8B, Llama3.1-8B, and Gemma3-12B using a corpus of 37 Indian legal acts to analyze their impact on retrieval and response generation.